# WATER : SCIENCE, SOCIETY & POLICY

ENVS/SDP 251 Spring Semester 2025

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# The Bernoulli Equation

$$P + \frac{1}{2}\rho v^2 + \rho g h = \text{constant}$$

https://www.shutterstock.com/search/bernoulli-equation

# BERNOULLI'S EQUATION

$$P + \frac{1}{2}\rho v^2 + \rho g h = \text{constant}$$

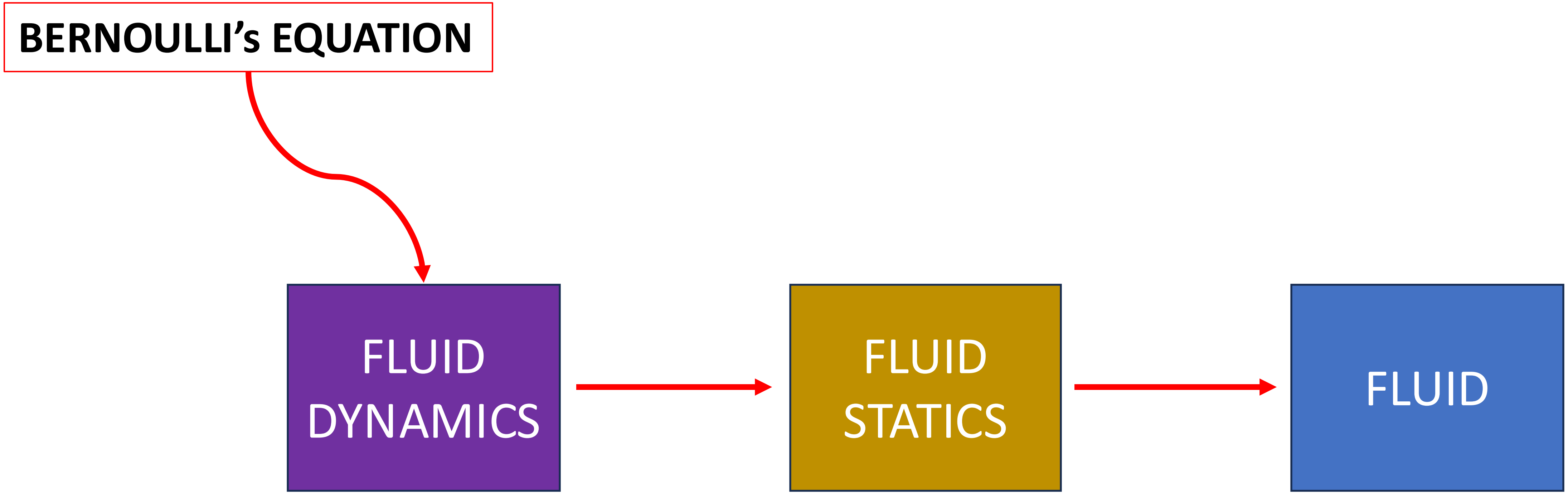
Pressure form

$$\frac{P}{\rho} + \frac{1}{2} v^2 + g h = \text{constant}$$

Energy form

$$\frac{P}{\rho g} + \frac{1}{2g} v^2 + h = \text{constant}$$

Head form


BERNOULLI's EQUATION
FLUID DYNAMICS
FLUID STATICS
FLUID

# WHAT IS A FLUID?

- Anything that can flow (also described as the ability to take on the shape of the container).

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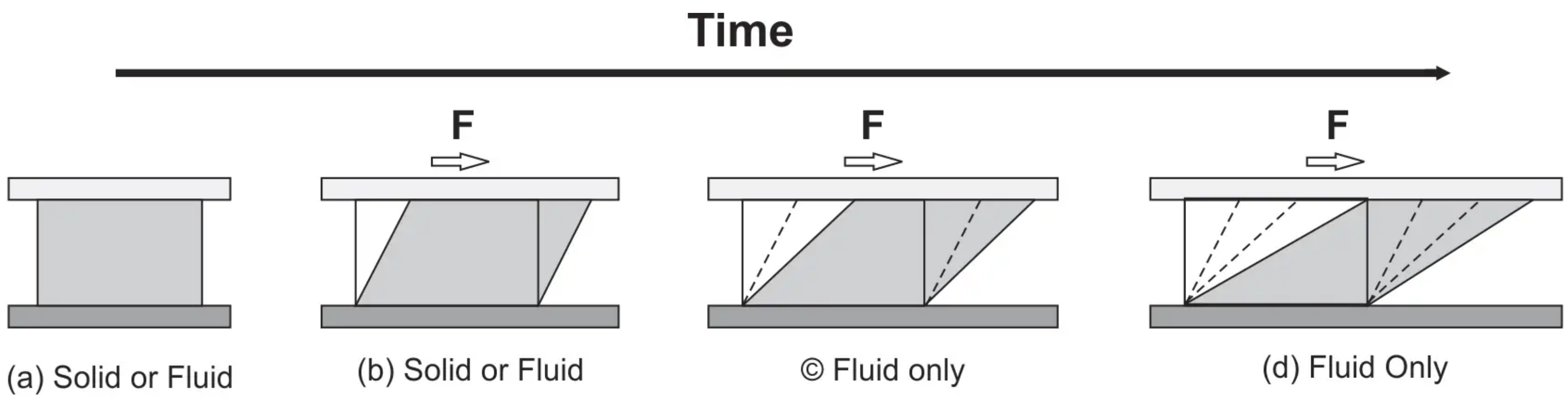
- A fluid is a material that continuously deforms (flows) under an applied shear stress, or external force.




https://www.mechanicalbooster.com/2021/03/types-of-fluid-flow.html/shear-stress-behaviour-of-solid-and-fluid

# BERNOULLI'S EQUATION

$$P + \frac{1}{2}\rho v^2 + \rho g h = \text{constant}$$

**Pressure form**

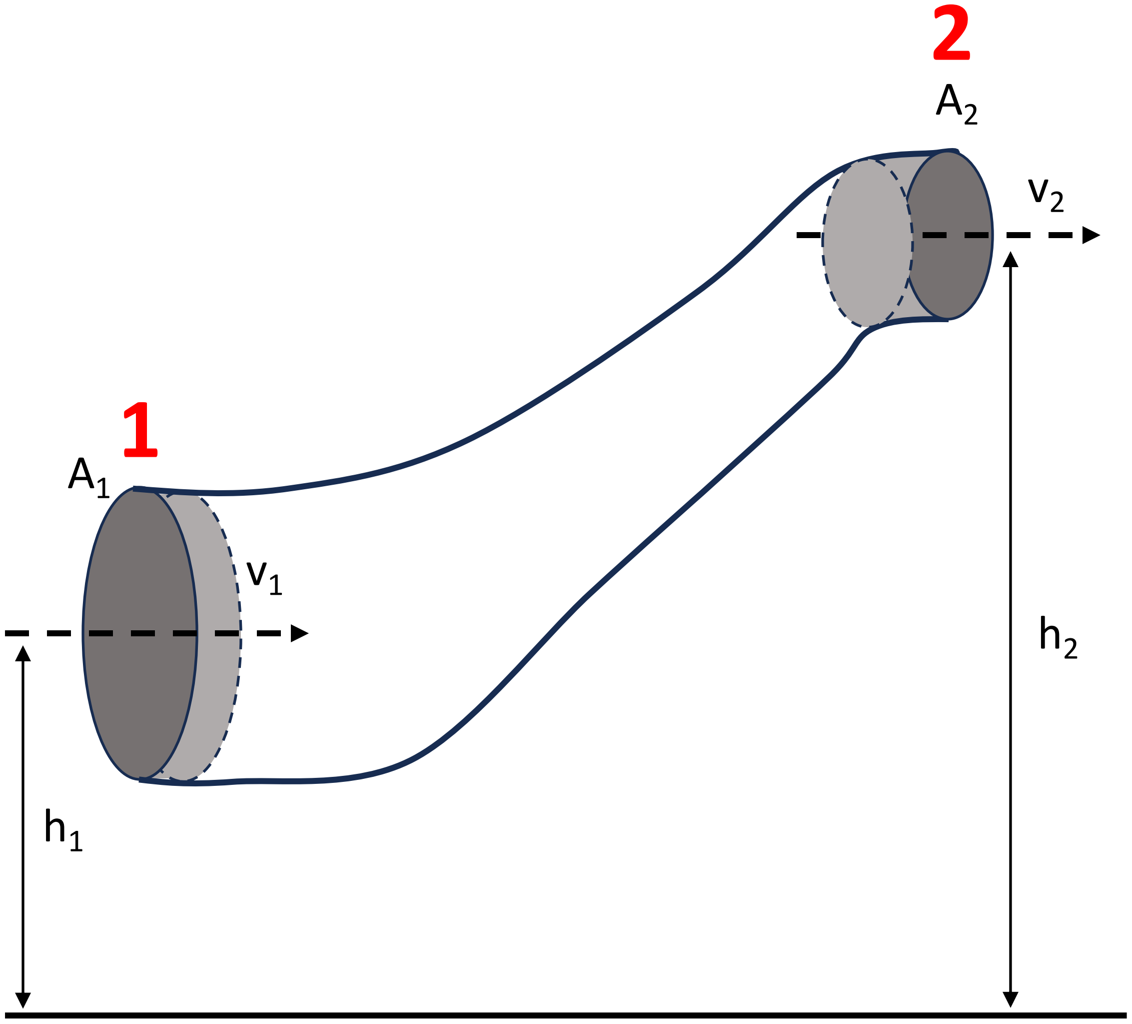
**For a fluid flowing between two points**

$$P_1 + \frac{1}{2}\rho {v_1}^2 + \rho g h_1 = P_2 + \frac{1}{2}\rho {v_2}^2 + \rho g h_2$$

# FLUID DYNAMICS (Fluids in Motion)

**Assumptions:**

- Steady State, no fluctuations over time
- Single input, single output
- Incompressible fluid
- Isothermal condition
- No phase change, no reaction
- No heat transfer
- Inviscid fluid



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**Continuity:**

$V_1 = V_2$

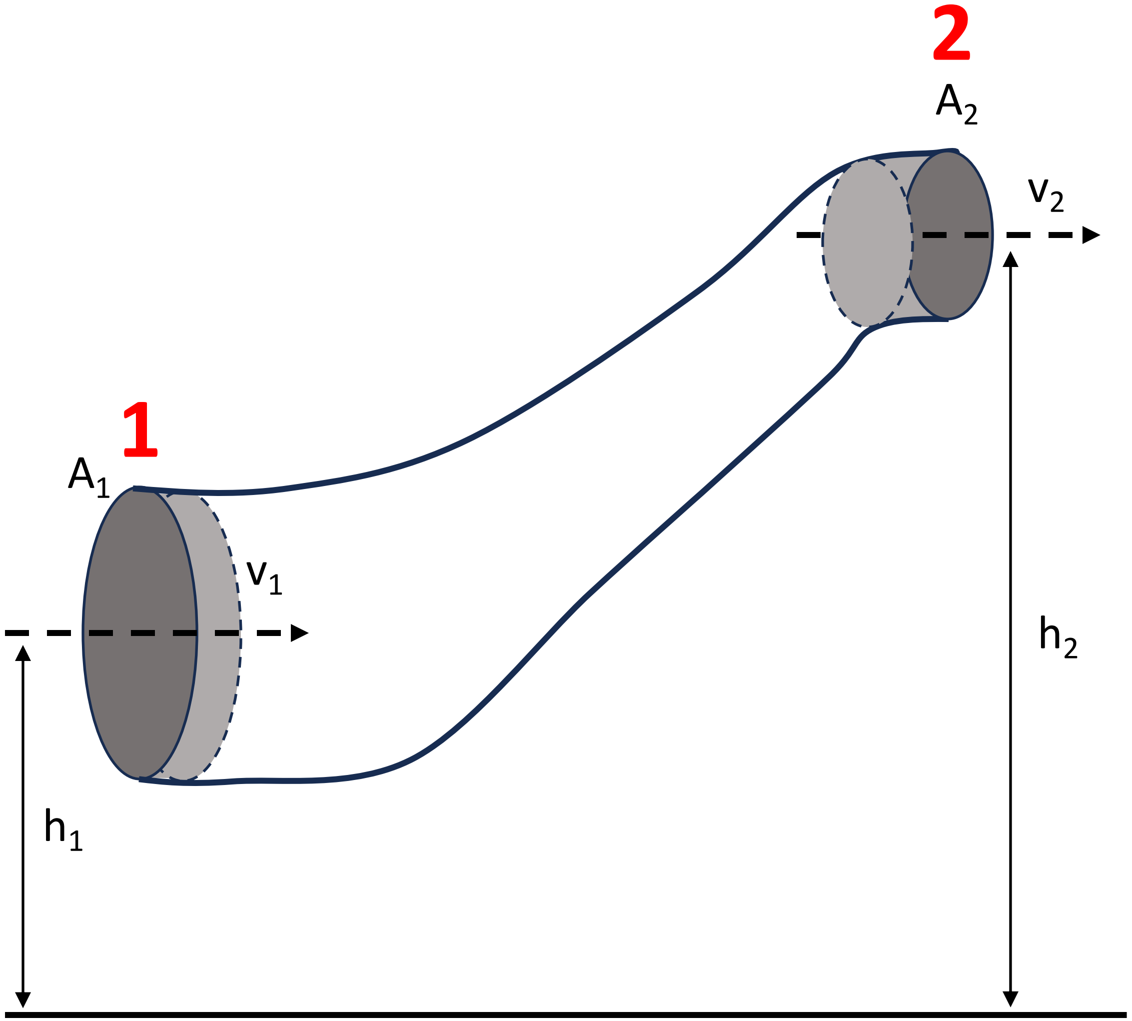
$A_1 \Delta x_1 = A_2 \Delta x_1$

$A_1(v_1\ \Delta t) = A_2(v_2\ \Delta t)$

$\boxed{A_1 v_1 = A_2 v_2}$

$\rho\ A_1 v_1 = \rho\ A_2 v_2$

$m_1 = m_2$



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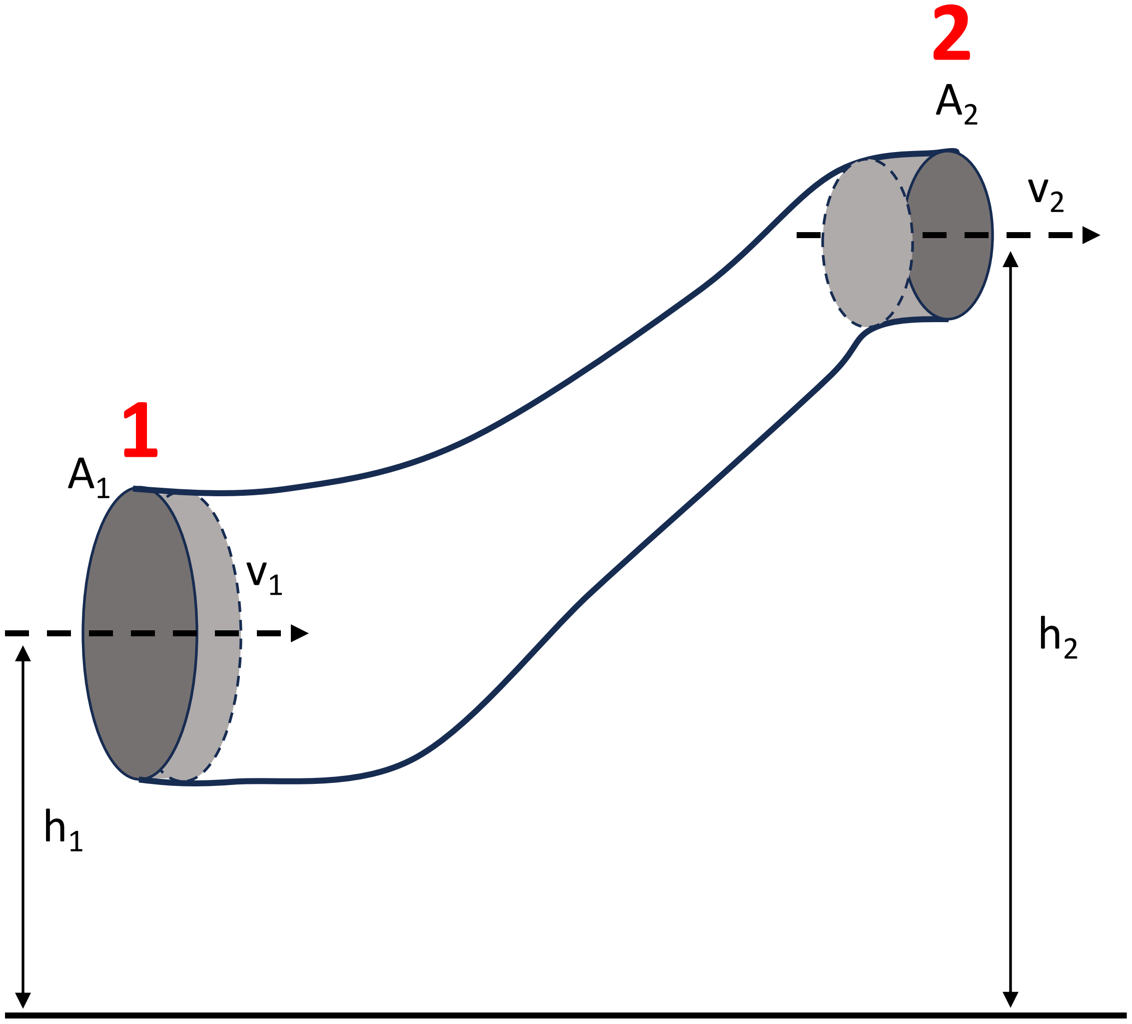
**Continuity:**

$A_1v_1 = A_2v_2$



# FLUID DYNAMICS

- If there are no external forces working on the system

  **K.E. + P.E. = K.E. + P.E.**

- If there are external forces:

  **$E_2 - E_1$ = Work done**

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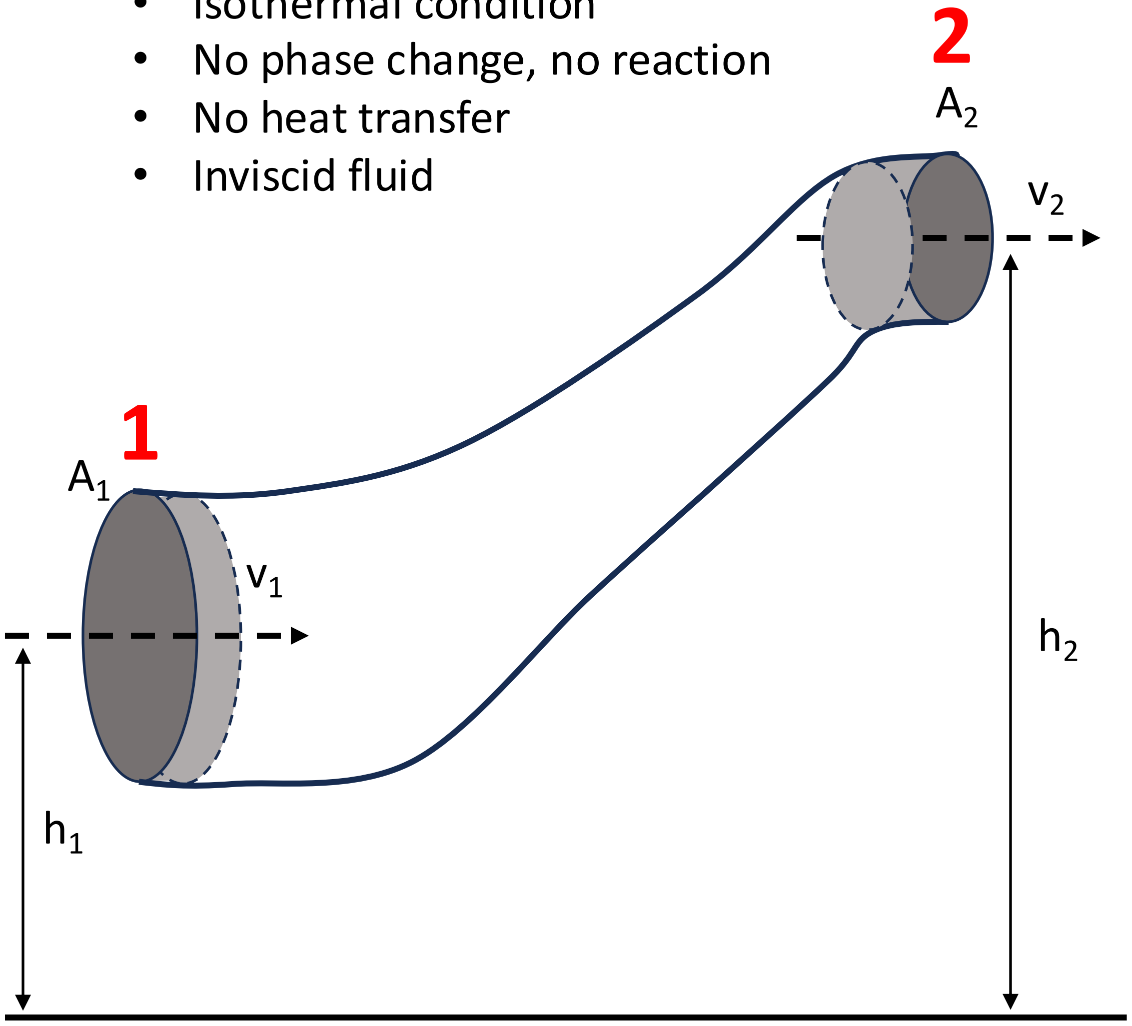

# FLUID DYNAMICS

- If there are no external forces working on the system

**K.E. + P.E. = K.E. + P.E.**

- If there are external forces:

**$E_2 - E_1$ = Work done (by external forces)**

Energy at point 2:

$$\rho A_2 \Delta x_2 \frac{v_2{}^2}{2} + \rho A_2 \Delta x_2 g h_2$$

Energy at point 1:

$$\rho A_1 \Delta x_1 \frac{v_2{}^2}{2} + \rho A_1 \Delta x_1 g h_1$$

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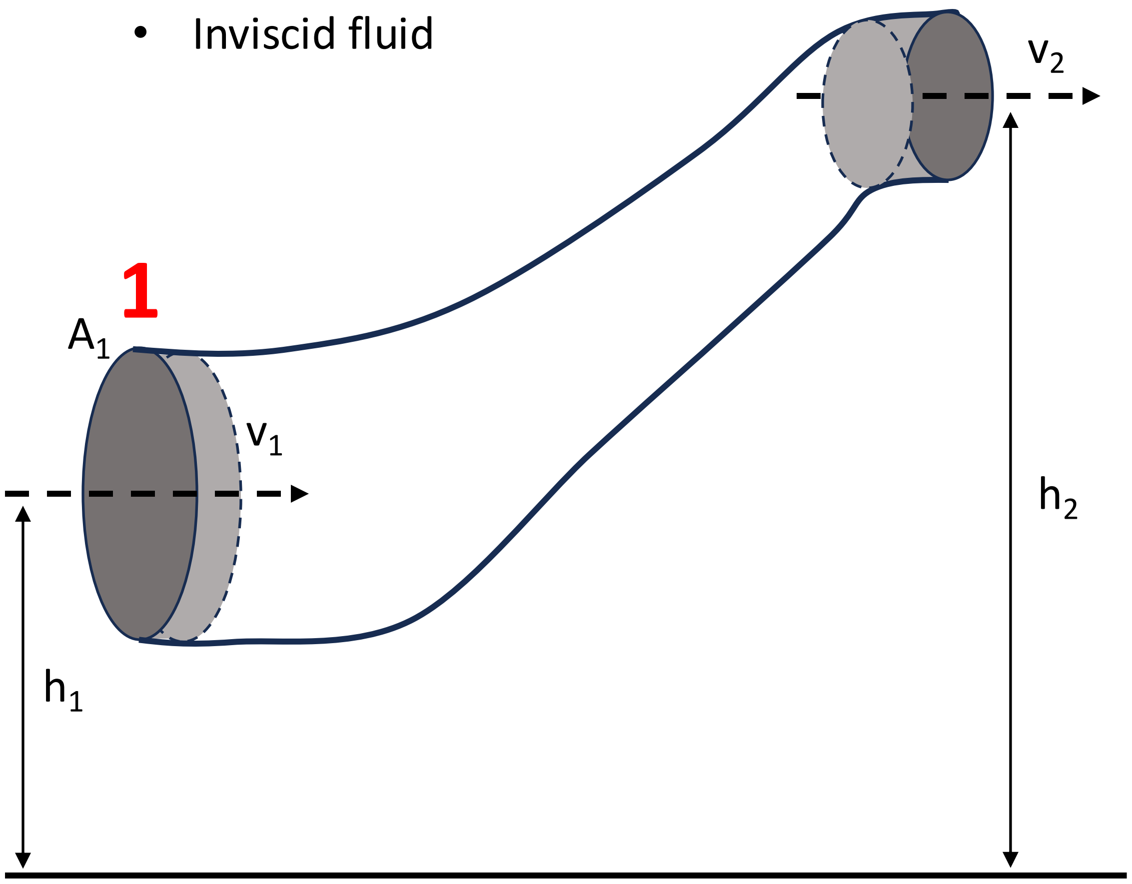

# FLUID DYNAMICS

$$E_2 - E_1 = W$$

$$\left(\rho A_2 \Delta x_2 \frac{v_2{}^2}{2} + \rho A_2 \Delta x_2 g h_2\right) - \left(\rho A_1 \Delta x_1 \frac{v_2{}^2}{2} + \rho A_1 \Delta x_1 g h_1\right)$$

$$= \left(P_1 A_1 \Delta x_1 - P_2 A_2 \Delta x_2\right)$$

$$\boxed{P_1 + \frac{1}{2}\rho v_1{}^2 + \rho g h_1 = P_2 + \frac{1}{2}\rho v_2{}^2 + \rho g h_2}$$

$$\boxed{P + \frac{1}{2}\rho v^2 + \rho g h = \text{constant}}$$

**Bernoulli's Equation**

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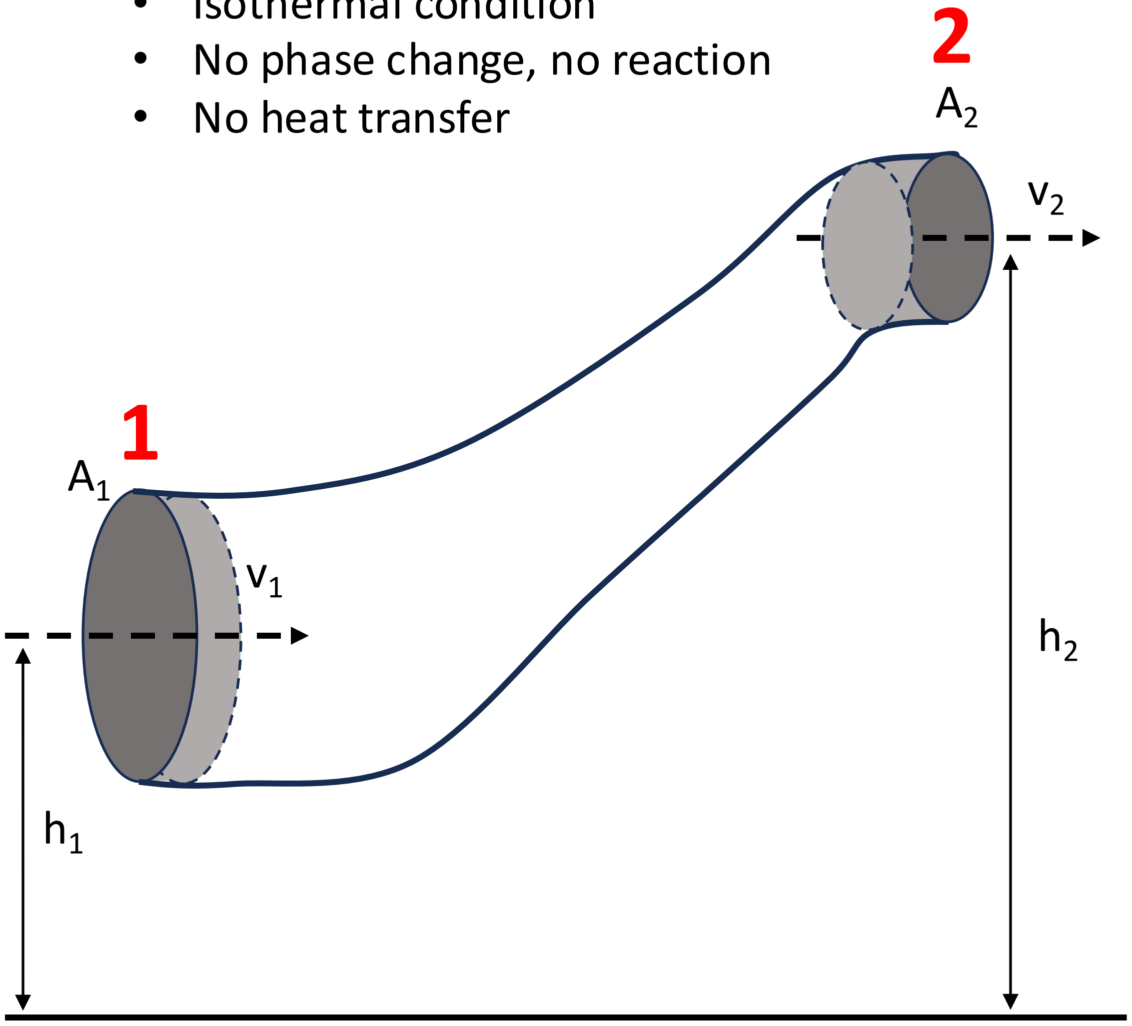
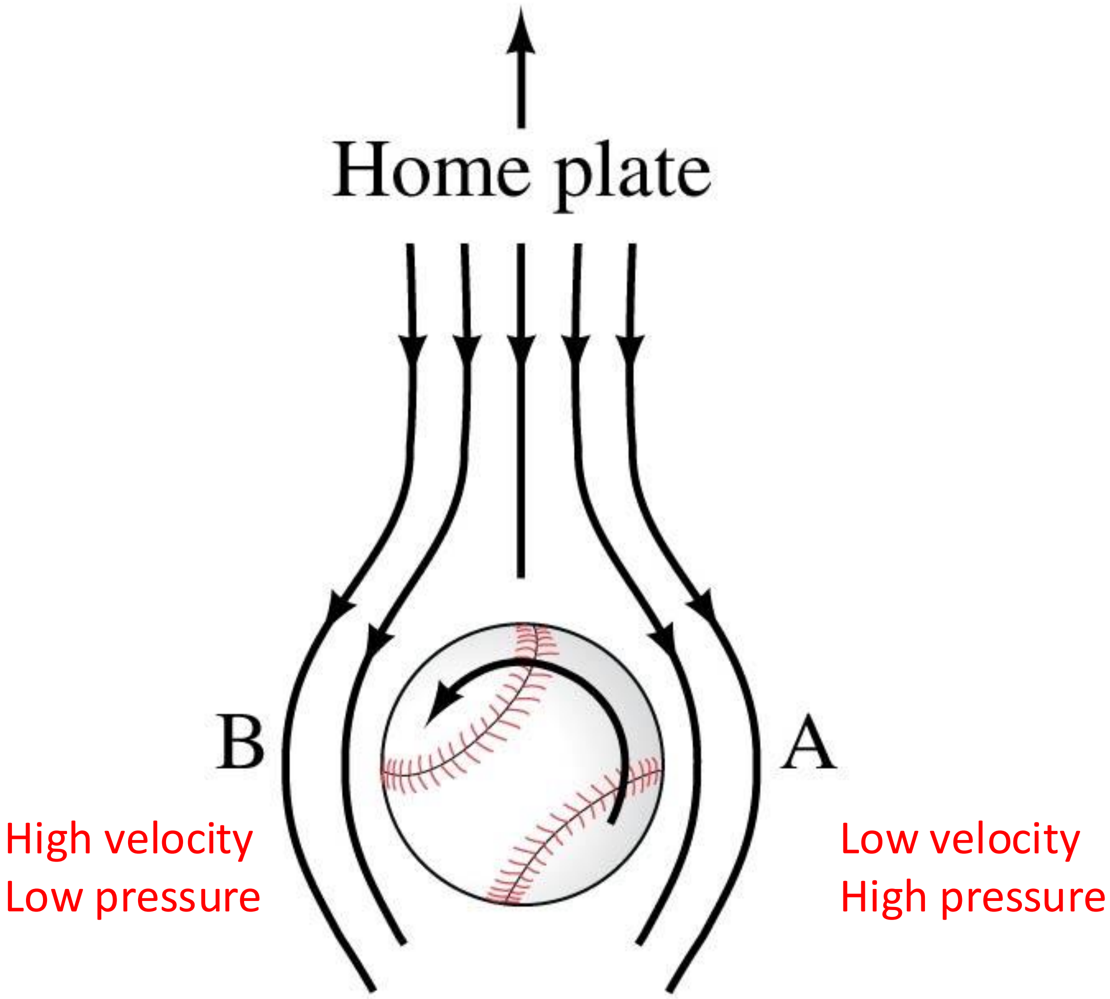


# APPLICATIONS OF BERNOULLI

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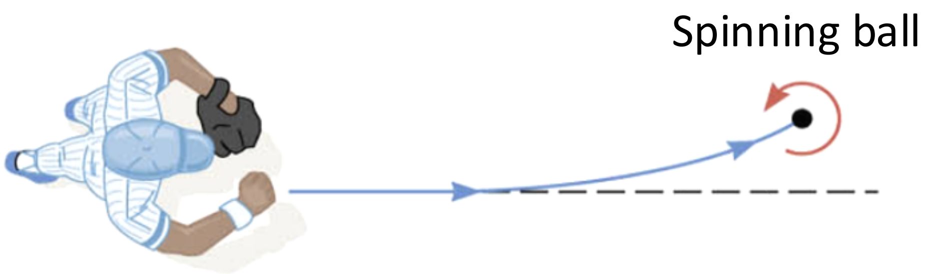
## Baseball spin

A ball's path will curve to left due to its spin, which results in the air speeds on the two sides of the ball not being equal.

$$P_1 + \frac{1}{2}\rho v_1^2 + \cancel{\rho g h_1} = P_2 + \frac{1}{2}\rho v_2^2 + \cancel{\rho g h_2}$$

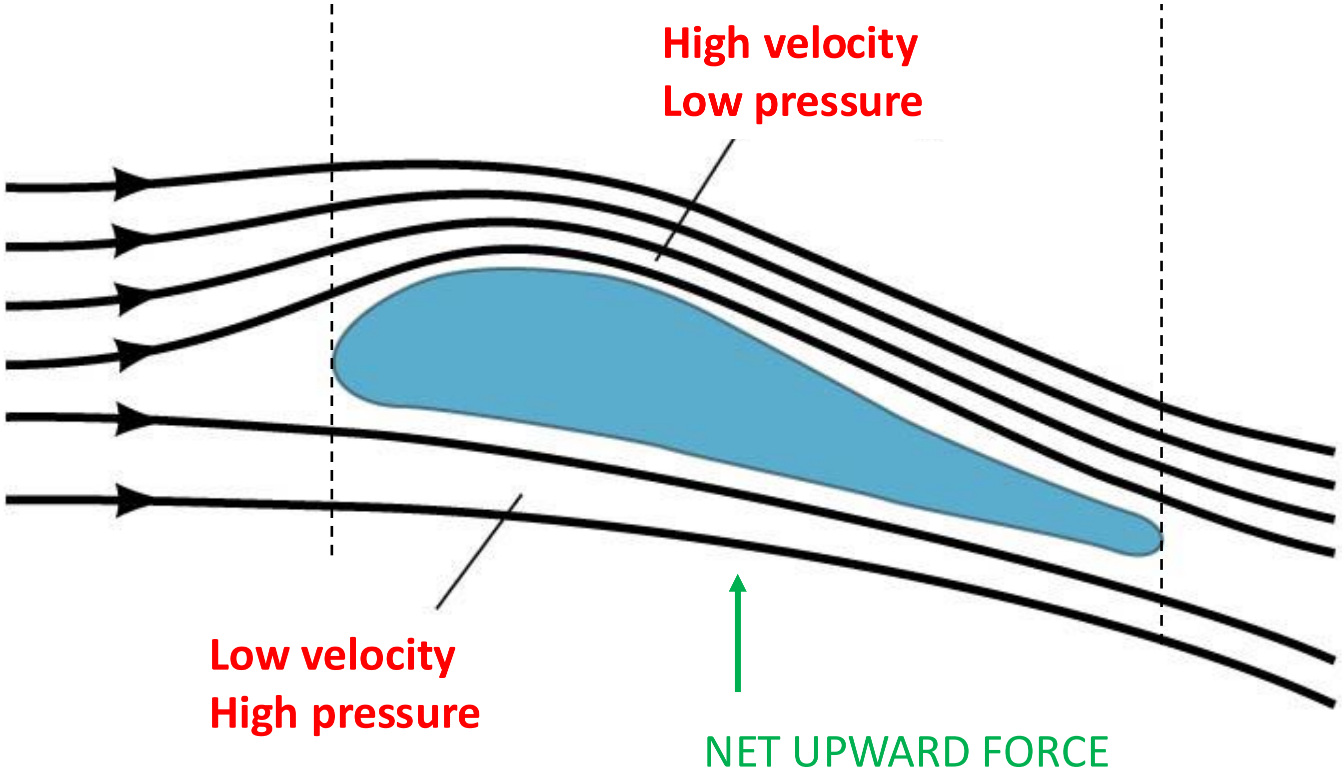
$$P_1 + \frac{1}{2}\rho v_1^2 = P_2 + \frac{1}{2}\rho v_2^2$$



# APPLICATIONS OF BERNOULLI

## Airplane lift

Lift on an airplane wing is due to the different air speeds and pressures on the two surfaces of the wing.

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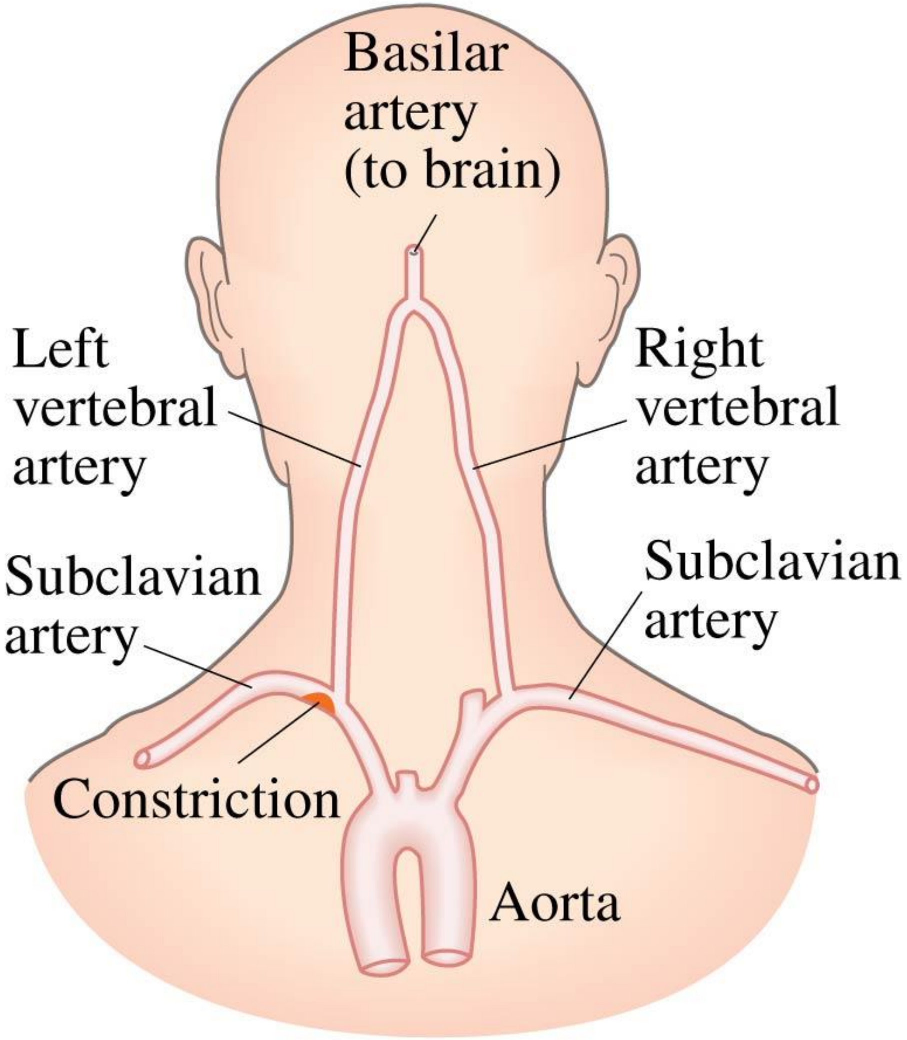


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## Constricted arteries

A person with constricted arteries will find that they may experience a temporary lack of blood to the brain as blood speeds up to get past the constriction, thereby reducing the pressure.

# Navier Stokes Equation

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- Inviscid fluid

$$\boxed{\Delta P + \frac{1}{2}\rho\Delta v^2 + \rho gh = \text{constant}} \Rightarrow \text{Bernoulli's Equation}$$

$$\boxed{\rho\frac{dv}{dt} = -\nabla p + \rho g} \Rightarrow \text{Euler's Equation}$$

$$\boxed{\rho\frac{dv}{dt} = -\nabla p + \mu\nabla^2 v + F} \Rightarrow \text{Navier Stokes Equation}$$